

A mathematical explanation of empirical results in machine learning

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Context

Modern machine learning achieves outstanding predictive performance, but classical statistical learning theory [5, 8] fails to explain many of its successes [2, 4].

Example

Generalization ability of over-parametrized models vs. classical overfitting arguments.

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★ We need new mathematical tools to explain successful results ★

Why should we bother?

- ▶ **scientific curiosity**
- ▶ **explainability**
- ▶ **future developments**

“nothing is more practical than a good theory” [7]

- SOTA: multiple currents, but no unified theory as broad as empirical practice

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Research question

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1. **mathematical framework** and hypotheses
2. test hypotheses with **experiments**
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4. if not: run **more complex experiments** to test robustness

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▶ *definition* $\rightarrow$ *theorem* $\rightarrow$ *proof* [9, 4]

▶ mental pictures

▶ just enough theory

Example: the puzzle of generalization

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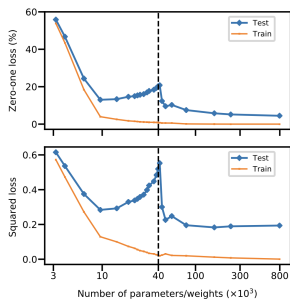
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In the context of gradient-based learning algorithms, **why do over-parametrized models generalize better?**

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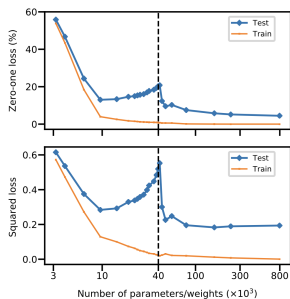


Double descent - Belkin et al. [3]

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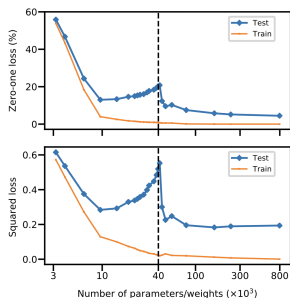
→ Analysis discrete update vs. continuous-time flow (**implicit regularization**)

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→ Analysis discrete update vs. continuous-time flow (**implicit regularization**)

- ▶ mathematical formalization of the problem
- ▶ experiments on gradient-based algorithms dynamics
- ▶ formulate and test hypotheses
- ▶ apply framework to test implicit regularization argument

Thank you

Questions?

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Appendix I

Proposition [Comparing discrete and continuous trajectories]
[1, 6]

For a step size $\alpha \in \mathbb{R}_{>0}$, $t \in \mathbb{R}_{>0}$

$$\theta_{t+\alpha} = \theta_t - \alpha \nabla \ell(\theta_t) \quad \text{and} \quad \frac{d\theta}{dt} = -\nabla \ell(\theta(t))$$

Up to order α , gradient descent corresponds to gradient flow of:

$$\ell_{discr=contin}(\theta) = \ell(\theta) + \frac{\alpha}{4} \|\nabla \ell(\theta)\|^2$$

Proof:

Consider the usual (discrete-time) update rule:

$$\theta_{\tau+1} = \theta_{\tau} - \alpha \nabla \ell(\theta_{\tau})$$

for $\tau \in \mathbb{N}$, $\alpha \in \mathbb{R}_{>0}$ step size.

Appendix II

Let $t = \alpha\tau$ and consider the finite-difference

$$\theta_{t+\alpha} - \theta_t = -\alpha \nabla \ell(\theta_t).$$

Now, taking the limit we obtain the continuous-time update rule (where we omit the dependence of θ on t to ease notation)

$$\lim_{\alpha \rightarrow 0} \frac{\theta_{t+\alpha} - \theta_t}{\alpha} = \frac{d\theta}{dt}.$$

Then,

$$\frac{d\theta}{dt} = -\nabla \ell(\theta). \quad (1)$$

Allowing

$$\frac{d\theta}{dt} = -\nabla \ell(\theta) + \alpha g(\theta),$$

we want to find the correction term g such that the continuous and discrete trajectories agree up to order α . Let us consider the

Appendix III

function $\theta(t)$ and inspect its Taylor expansion starting from $\theta_0 = \theta(0)$, for a jump $\Delta t = \alpha$ (1 iteration in discrete sense):

$$\begin{aligned}\theta(t) &\approx \theta_0 + \alpha \frac{d\theta}{dt} + \frac{1}{2} \alpha^2 \frac{d^2\theta}{dt^2} \\ &= \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 g(\theta) + \frac{1}{2} \alpha^2 \frac{d}{dt} \left(-\nabla \ell(\theta) + \alpha g(\theta) \right) \\ &\approx \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 \left(g(\theta) + \frac{1}{2} \left(-\nabla^2 \ell(\theta) \frac{d\theta}{dt} \right) \right) \\ &= \theta_0 - \alpha \nabla \ell(\theta) + \alpha^2 \left(g(\theta) + \frac{1}{2} \left(\nabla^2 \ell(\theta) \left(\nabla \ell(\theta) + \alpha g(\theta) \right) \right) \right)\end{aligned}$$

where the terms in pink are of 3rd order in α , hence disregarded. Then, we impose $\theta(t) = \theta_{\tau+1}$, obtaining

$$g(\theta) = -\frac{1}{2} \nabla^2 \ell(\theta) \nabla \ell(\theta).$$

Appendix IV

Hence, the following continuous-time update rule agrees with the discrete-time update up to order α

$$\frac{d\theta}{dt} = -\nabla\ell(\theta) - \frac{1}{2}\alpha\nabla^2\ell(\theta)\nabla\ell(\theta). \quad (2)$$

Lastly, we notice that this is the same as applying the usual continuous-time update rule to the modified loss

$$\tilde{\ell}(\theta) = \ell(\theta) + \frac{1}{4}\alpha\nabla\ell(\theta)^t\nabla\ell(\theta) = \ell(\theta) + \frac{1}{4}\alpha\|\nabla\ell(\theta)\|^2 \quad (3)$$